ANUSHKA GUPTA

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Computer Science undergraduate with software engineering and AI/Computer Vision internship experience at CRIS and L2M Rail. Strong foundation in Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, and Computer Networks. Skilled in developing full-stack web applications and AI-powered computer vision solutions using modern software engineering practices.

Education

Banasthali Vidyapith, Bachelor of Technology (2023-2027) – Computer Science Engineering **(VII Semester)**
CGPA: 8.49/10
Banasthali Vidyapith, 2022 - 2023 ● *Banasthali (Class XII), Aggregate: 82%*
Banasthali Vidyapith, 2019 - 2022 ● *Banasthali (Class X), Aggregate: 93%*

Technical Skills

Programming Languages: C++, Python, Java, JavaScript, TypeScript, C, SQL
Core CS: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems (OS), Computer Networks (CN), RESTful API Design
Web Technologies: React, Node.js, Express.js, HTML5, CSS3
Databases: MongoDB, MySQL
AI & Libraries: PyTorch, OpenCV, YOLOv8, NumPy, Pandas
Tools: Git, GitHub, Linux, Bash, Postman

Work Experience

L2M Rail | AI / Computer Vision Intern — **July 2026 – Present**

- Engineered and evaluated deep learning architectures (PyTorch, ResNet) for real-time automated railway component inspection.
- Performed dataset curation, false-positive analysis, and model evaluation to improve railway defect detection reliability.
- Developed scalable computer vision pipelines for processing large-scale railway inspection datasets.

Centre for Railway Information Systems (CRIS) | Software Intern — **June 2024 - July 2024**

- Developed 6 production-oriented web applications using Django and MySQL to automate internal administrative workflows.
- Designed normalized MySQL database schemas and optimized backend queries for improved performance.
- Implemented backend validation, secure routing, and exception handling following software engineering best practices.

Projects

Alumni Connect Platform | React, TypeScript, Node.js, Express, MongoDB — **Feb 2026**
https://github.com/anushkalearn/alumni-connect-portal

- Architected a full-stack web application using React, TypeScript, Node.js, Express, and MongoDB for multi-role user management (Students, Alumni, Admin).
- Implemented secure JWT-based authentication and role-based access control (RBAC).
- Designed RESTful endpoints supporting job referrals, mentorship booking, and real-time query routing.

CPU Scheduling Simulator | C++, Operating Systems — **Dec 2025**
https://github.com/anushkalearn/cpu-scheduling-simulator

- Implemented a modular C++ simulator executing FCFS, SJF, Priority, and Round Robin CPU scheduling algorithms.
- Calculated turnaround time, waiting time, and CPU utilization metrics to analyze algorithm efficiency.
- Built Gantt chart visualization logic in terminal output to trace real-time process execution timelines.

Personal Portfolio Website | HTML, CSS, JavaScript — **May 2025**
https://github.com/anushkalearn/personal-portfolio

- Designed and deployed a responsive developer portfolio using semantic HTML, modular CSS, and JavaScript to showcase projects, skills, and experience.
- Integrated a serverless contact workflow using Formspree API enabling recruiter communication.
- Live: https://anushkalearn.github.io/personal-portfolio/

Automatic Number Plate Recognition (ANPR) System | Python, OpenCV — **Oct 2025**

- Implemented template matching and image preprocessing algorithms (grayscale, thresholding, edge detection) using Python and OpenCV to extract vehicular license plates.
- Developed segmentation techniques to isolate and recognize alphanumeric characters under variable lighting

conditions.